2026 H2 Physics Preliminary Examination Solutions
Paper 2

1 (a) (i) As the velocity of the ball increases from $t = 0$ to $t = 8.0$ s, the acceleration of the ball given by the gradient, $\frac{dv}{dt}$ decreases which implies that resultant force on the ball decreases.

The resultant force on the ball as it is falling from rest is given by the vector sum of the drag force F_D acting on the ball and its weight W i.e. $F_{resultant} = W - F_D$. Since W is constant, the drag force acting on the ball increases.

(ii) $$a = \frac{20.50 - 12.00}{6.0 - 1.0} = 1.70 \text{ m s}^{-2}$$

$$mg - F_D = ma$$
$$(0.80)(9.81) - F_D = (0.80)(1.70)$$
$$F_D = 6.488 = 6.5 \text{ N}$$

(b) (i) $$\eta = \frac{F_D}{6\pi rv}$$

base units of $F_D = \text{kg} \times (\text{m s}^{-2})$

base units of $\eta = \frac{\text{kg} \times (\text{m s}^{-2})}{\text{m} \times (\text{m s}^{-1})}$
$= \text{kg m}^{-1} \text{ s}^{-1}$

(ii) The maximum drag force will be larger as it is always equal to the weight of the ball (at terminal velocity).

2 **(a)** The waves have a <u>constant phase difference</u>.

(not accepted: same amplitude, wavelength, velocity)

(b)

$$x = \frac{\lambda D}{a}$$

$$= \frac{530 \times 10^{-9} (2.7)}{0.85 \times 10^{-3}}$$

$$= 1.6835 \times 10^{-3} = 1.7 \times 10^{-3} \text{ m}$$

(c) **(i)** Less light passes through the narrower slits resulting in <u>dimmer bright fringes</u>. There is no change in the fringe separation as slit width is unchanged.

(note: 'decrease in contrast' is ambiguous. It must be clear that bright fringes are dimmer only. Dark fringes remain dark.)

(ii) As the screen is rotated, the distance between different parts of the screen to the double slit will not be the same.

The <u>fringe separations</u> will be <u>shorter</u> when the screen is <u>nearer</u> to the double slit, and <u>longer</u> on the part of the screen <u>further</u> away (fringe separation is proportional to slits-to-screen distance).

The fringes <u>when the screen is nearer to the double slits are brighter</u> and the fringes when the screen is <u>further away are dimmer</u> (light is spread over a larger area when the screen is further away).

3 **(a)** Using $pV = nRT$,

$$n = \frac{pV}{RT} = \frac{1.0 \times 10^5 \times 6.0 \times 10^{-4}}{8.31 \times 300}$$
$$= 0.024067 = 0.024 \text{ mol}$$

(b) Using the ideal gas equation $pV = nRT \Rightarrow n = \frac{pV}{RT}$.
For a fixed mass of gas, n is the same at both the initial state (1) and final state (2).

$$\therefore \frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$
$$p_2 = \frac{p_1V_1T_2}{T_1V_2} = \frac{1.0 \times 10^5 \times 6.0 \times 10^{-4} \times 550}{300 \times 5.0 \times 10^{-5}}$$
$$= 2.2000 \times 10^6 = 2.2 \times 10^6 \text{ Pa}$$

Alternatively using state (1),

$$p_2 = \frac{nRT_2}{V_2} = \frac{0.024067 \times 8.31 \times 550}{5.0 \times 10^{-5}}$$
$$= 2.19996 \times 10^6 = 2.2 \times 10^6 \text{ Pa}$$

(c) Since the gas can be treated as an ideal gas, increase in internal energy ΔU is given by

$$\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} \times 0.024 \times 8.31 \times (550 - 300)$$
$$= 74.79 = 75 \text{ J}$$

Alternatively,

$$\Delta U = \frac{3}{2} \Delta(pV) = \frac{3}{2} \times \left[(p_2V_2 - p_1V_1) \right]$$
$$= \frac{3}{2} \times \left[(2.2 \times 10^6 \times 5.0 \times 10^{-5}) - (1.0 \times 10^5 \times 6.0 \times 10^{-4}) \right]$$
$$= 75 \text{ J}$$

(d) **(i)** The first law of thermodynamics states that the <u>increase</u> in the <u>internal energy</u> of a system is <u>equal</u> to the <u>sum</u> of the <u>heat supplied to</u> the system and the <u>work done on</u> the system, and the internal energy of a system depends only on its state.

(ii) The <u>gas loses heat</u> hence Q is <u>negative</u>. Compression stroke is rapid to <u>minimise heat loss from the gas</u> to the engine block/surroundings, so that Q is less negative.

Since <u>gas</u> is <u>compressed</u>, work done on it W <u>is positive</u>. From first law of

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thermodynamics $\Delta U = W + Q$. If Q is less negative, the gain in internal ΔU is maximised.

For ideal gas, $\Delta U \propto \Delta T$ where ΔT is the temperature increase during the compression stroke and ΔT will correspondingly be maximised. This leads to higher efficiency for the engine.
OR

When the compression stroke takes place rapidly, heat loss from the gas to the surroundings is minimised such that Q can be assumed to be 0.

Since gas is compressed, work done on it W is positive. From first law of thermodynamics $\Delta U = W + Q$. Since $Q = 0$ (and not negative i.e. there is heat loss from the gas), the gain in internal ΔU is maximised.

For ideal gas, $\Delta U \propto \Delta T$ where ΔT is the temperature increase during the compression stroke and ΔT will correspondingly be maximised. This leads to higher efficiency for the engine.

4 **(a)** **(i)** The charge is positive as it is attracted to the negative plate (or repelled by the positive plate).

(ii) Since angle is 45°,

$F_E = mg$

$= (0.25 \times 10^{-3})(9.81)$

$= 2.4525 \times 10^{-3} = 2.5 \times 10^{-3}$ N

(b) $\tan\theta = \frac{F_E}{mg} = \frac{Q\,\Delta V}{d\,mg}$

graph of $\tan\theta$ against $\frac{1}{d}$ gives straight line through origin with gradient $= \frac{Q\,\Delta V}{mg}$

From graph, gradient $= \frac{150.0}{40.0}$

Hence $\frac{Q\,\Delta V}{mg} = \frac{150.0}{40.0}$

$Q = \frac{150.0}{40.0}\left(\frac{(0.25 \times 10^{-3})(9.81)}{12}\right)$

$= 7.6641 \times 10^{-4} = 7.7 \times 10^{-4}$ C

(c) When equilibrium is reached, $V_1 = V_2$

$\frac{Q_1}{C_1} = \frac{Q_2}{C_2}$

$Q_1 = \frac{Q_2}{C_2} C_1 = \frac{Q_2}{10} \times 5.0$ = 0.50 Q_2 ……. (1)

Charges are conserved,

$Q_1 + Q_2 = C_1V_1 + C_2V_2 = (5.0 \times 10^{-12})(12.0) + (10.0 \times 10^{-12})(6.0) = 1.2 \times 10^{-10}$ C …… (2)

Solving (1) and (2):

$Q_1 = 4.0 \times 10^{-11}$ C

$Q_2 = 8.0 \times 10^{-11}$ C

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5 **(a)** As the triangular loop exits the field, the area of the loop remaining inside the field decreases.

The magnetic flux through the loop decreases and by Faraday's law, an e.m.f. is induced in the loop.

(b) Once the triangle starts to exit the field, the area within the magnetic field decreases by a triangular segment where the base (vt) and height (which is the same as the base) are both proportional to time.

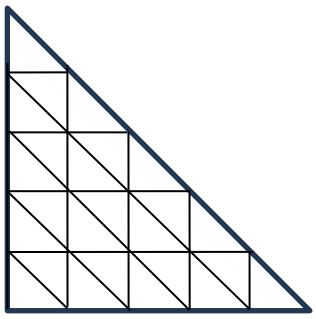
$$\therefore A = A_0 - \frac{1}{2}(vt)(vt) = A_0 - \frac{1}{2}v^2t^2$$

$$E = \frac{d(BA)}{dt} = \frac{Bd\left(A_0 - \frac{1}{2}v^2t^2\right)}{dt}$$

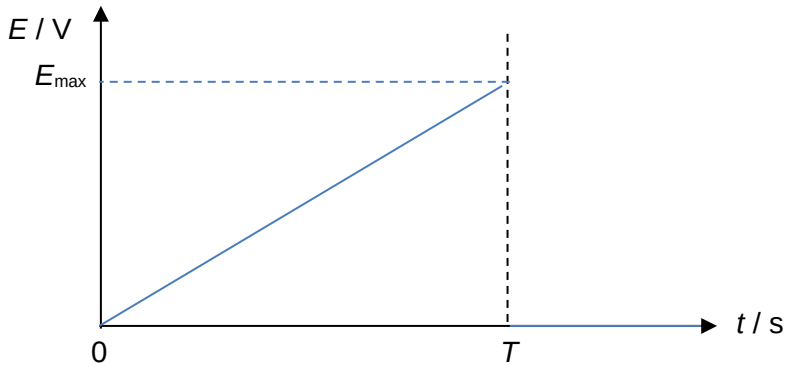
$$= Bv^2t$$

$$\Rightarrow E \propto \frac{dA}{dt} \propto t$$

OR
In the first Δt, area decreases by 1 small triangle.
In the second Δt, area decreases by 3 small triangles.
In the third Δt, area decreases by 5 small triangles.
In the fourth Δt, area decreases by 7 small triangles.



Hence, the rate of decrease of area increases at a constant rate and E increases at a constant rate.



*E increasing linearly until T

(c)

Total change in area $= \frac{1}{2}(0.15)^2 = 0.01125 \text{ m}^2$

Total time taken $= \frac{0.15}{0.30} = 0.50 \text{ s}$

$$E_{ave} = \frac{\Delta\phi}{\Delta t} = \frac{B\Delta A}{\Delta t} = \frac{(0.40)(0.01125)}{0.50}$$
$$= 0.0090 \text{ V}$$

6 **(a)** No photoelectrons are emitted (from the metal surface) below the threshold frequency, f_0 regardless of the light intensity.

Electromagnetic radiation consists of photons with energy $E = hf$ in a 1-to-1 interaction with electrons, so a single photon with energy below hf_0 lacks the minimum energy required to free an electron.

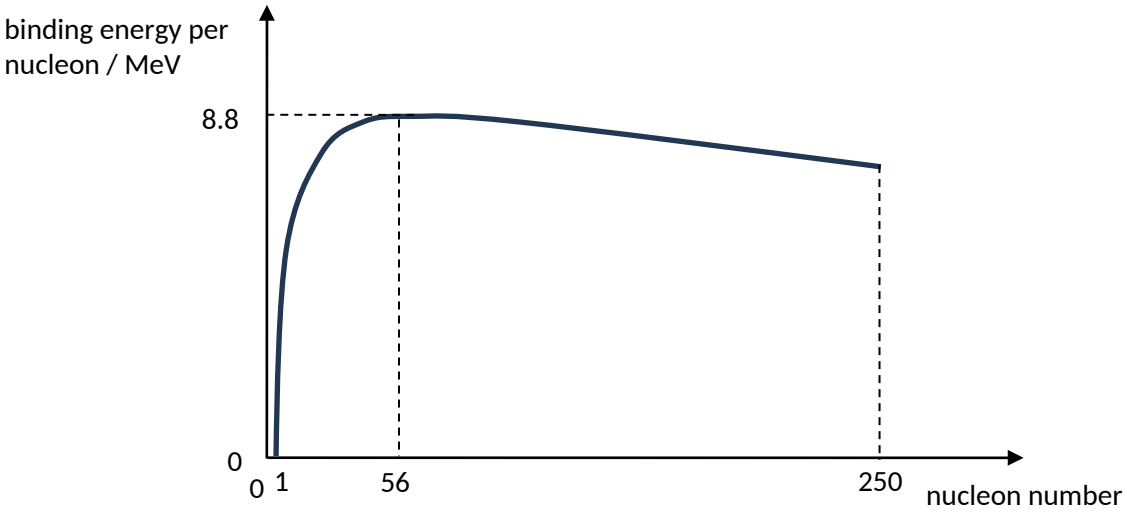
(b) The emission line spectrum consists of a series of distinct lines of specific colours/ wavelengths/ frequencies on a dark background.

Each line is produced by the emission of photons of a specific quantum of energy. Since the photons have specific energy, they must be emitted when electrons in the atoms de-excite from a higher to lower discrete energy level.

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7 **(a)** The binding energy of a nucleus is the <u>energy required to separate to infinity all the nucleons of a nucleus.</u>

(b)



*correct shape of curve, nucleon numbers 1 and 250 and highest point labelled.

(c) The lighter product (daughter) nuclei have higher binding energy per nucleon than the "parent" nucleus.
This means that the formation of the product (daughter) nuclei from their constituent nucleons releases more energy than that required to completely disintegrate the "parent" nucleus.

(d) **(i)** number of α-particles = 0
number of β-particles = 2

(ii) It is not feasible as a tremendous amount of energy is required to overcome the <u>strong electrostatic (Coulomb) repulsion</u> between the nuclei.

(e) decrease in mass = 233.039634 - 136.911557 - 93.915355 - 2(1.008664)
= 0.195394 u

energy $= \Delta mc^2$

$= 0.195394 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$

$= 2.919186 \times 10^{-11} = 2.92 \times 10^{-11}$ J

8 (a) (i) A petrol vehicle consumes 7.0 litres of fuel per 100 km and 1 litre of petrol has an energy of 34.2 MJ.

$$P = \frac{E}{t}$$
$$= \frac{7.0(34.2 \times 10^6)}{60 \times 60}$$
$$= 66500 \text{ W} = 67 \text{ kW}$$

(ii) energy per km used by petrol vehicle
$$= \frac{7.0(34.2 \times 10^6)}{100}$$
$$= 2.394 = 2.4 \text{ MJ km}^{-1}$$

energy per km used by EV
$$= \frac{(18 \times 10^3) \times 60 \times 60}{100}$$
$$= 0.648 \text{ MJ km}^{-1}$$

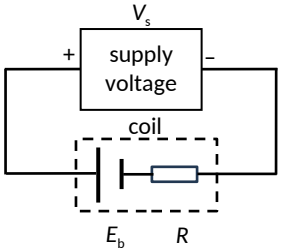
energy per km generated by power plant for EV
$$= 2 \times 0.648$$
$$= 1.296 = 1.3 \text{ MJ km}^{-1}$$

Even with energy losses at the power plant resulting in 50 % efficiency, the EV consumes nearly half the energy per km compared to a petrol vehicle and does not produce emissions when running. Hence the EV is considered 'greener'.

(b) (i) As the motor spins faster, the rate of cutting of magnetic field lines by the rotor coils increases. By Faraday's Law, this increases the induced back e.m.f.

According to Lenz's Law, this induced back e.m.f. acts in the direction to oppose the supply voltage. The net e.m.f. (voltage) driving the motor decreases, reducing the total current in the motor.

(ii) $$I = \frac{V_s - E_b}{R} = \frac{360 - 300}{0.60} = 100 \text{ A}$$



$$P_{mech} = IE_b = 100 \times 300 = 30000 = 30 \text{ kW}$$

OR

$$P_{mech} = P_{supply} - P_{loss}$$
$$= IV_s - I^2R$$
$$= (100)(360) - (100^2)(0.60)$$
$$= 30 \text{ kW}$$

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Note:

$P_{mech} = P_{supply} - P_{loss} = IV_s - I^2R = I(V_s - IR) = IE_b$

power supply = mechanical power developed in motor + power loss

The rate at which electrical energy is successfully converted into mechanical power (torque and rotational kinetic energy) is from doing work against the induced back e.m.f.

(c) (i) Since the ultracapacitor module is uncharged at $t = 0$ s,

$V_T = Ir$

$r = \frac{V_T}{I} = \frac{0.3}{120} = 0.0025\ \Omega$

(Actual value: $V_T = 0.31$ V at $t = 0$ s)

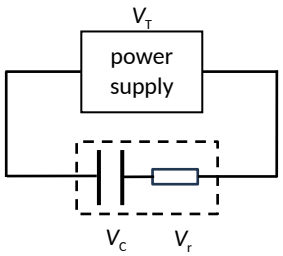
(ii) At $t = 40$ s, $V_T = 10.4$ V

$V_T = V_C + Ir$

$V_C = V_T - Ir$

$= 10.4 - (120)(0.0025)$

$= 10.1 = 10$ V



(iii) charging capacitor with constant current, $V_C = \frac{Q}{C} = \frac{I}{C}t$

with internal resistance, $V_T = \frac{I}{C}t + V_r$

using coordinates (13.0 , 3.6) and (46.0 , 12.0)

$\text{gradient} = \frac{12.0 - 3.6}{46.0 - 13.0}$

$\frac{I}{C} = 0.25455$

$C = \frac{120}{0.25455} = 471.42 = 470$ F

(d) (i) $N = \frac{42}{2.8} = 15$

(ii) effective capacitance for capacitors in series

$= \frac{3600}{15}$

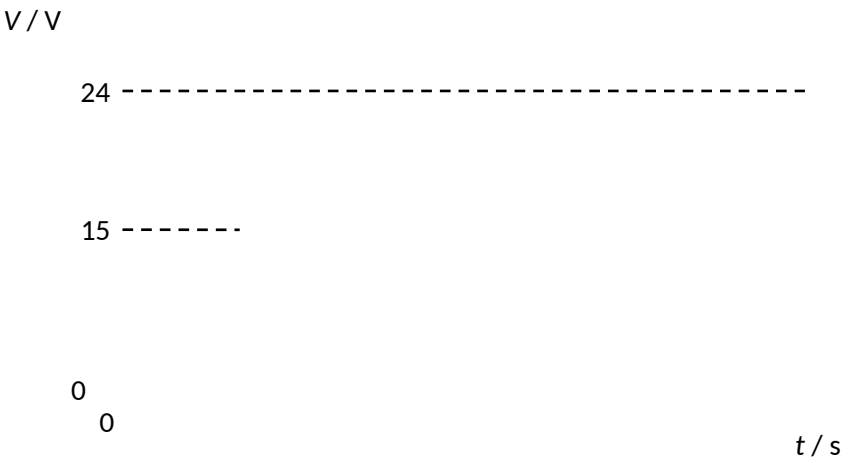
$= 240$ F

(iii) The <u>experimental module capacitance</u> of 470 F <u>is higher than a single series string</u> of 240 F. This implies that the module consists of <u>multiple series strings connected in a parallel arrangement</u>.

(e) (i) $\tau = 0.50 \times 471 = 235.5 = 240$ s

If $C = 470$, $\tau = 0.50 \times 471 = 235 = 240$ s

(ii)



when $t = \tau$, $V = 24\left(1 - e^{-1}\right) = 15.171 = 15 \text{ V}$

when $t = 5\tau$, $V = 24\left(1 - e^{-5}\right) = 23.838 = 24 \text{ V}$

*correct shape, smooth curve and correct value of 15 V at τ and approaching 24 V or 23.8 V at 5τ

(iii) **1.** total energy stored in module

$$= \frac{1}{2}CV^2$$
$$= \frac{1}{2}(471)\left(24^2\right)$$
$$= 1.35648 \times 10^5 = 1.4 \times 10^5 \text{ J}$$

2. total energy delivered by supply

$$= QV$$
$$= (CV)V$$
$$= (471)\left(24^2\right)$$
$$= 2.71296 \times 10^5 = 2.7 \times 10^5 \text{ J}$$

3. $\frac{E_{\text{stored}}}{E_{\text{supply}}} = \frac{1.35648 \times 10^5}{2.71296 \times 10^5} = 0.50$

The total energy stored in the module is half the energy delivered by the supply. Half of the energy delivered is dissipated as thermal energy in the series resistor / climate control load during charging.

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